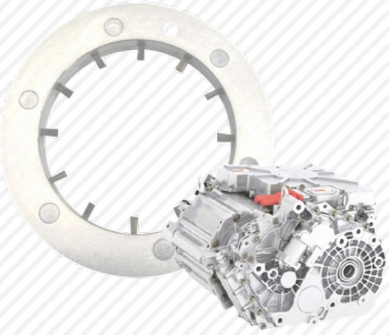
Material Innovation For a Better Life

2026

Long-term Solution for Bearing Electrical Corrosion Issue

Volsun Shaft Grounding Ring/ Conductive Brush Catalog

Suzhou Volsun Electronics Technology Co., Ltd.
苏州沃尔兴电子科技有限公司



About Volsun®

Volsun, founded in 2006, is a high-tech enterprise specializing in the research, development, production, and sales of polymer materials. With two major production bases in Yangzhou and Suzhou covering over 25,000 square meters, we provide a comprehensive range of safety protection solutions, including insulation, sealing, waterproofing, thermal conductivity, fireproofing, and electrical conductivity, widely applied across industries such as electricity, telecommunications, automotive, and electronics, etc.

Driven by innovation, Volsun has established strong industry-academia-research partnerships with leading Chinese universities and holds 97 technology patents, reinforcing our leadership in polymer materials technology.

Our core products include cold shrink/heat shrink tubes, fiberglass sleeves, silicone self-fusing tape, silicone overhead line cover, thermal pads, **conductive rings/brushes (CRs/CBs)**, and various insulation products.

As a technology-driven company, Volsun has been recognized as a National High-tech Enterprise and "Specialized, Refined, and Innovative" (SRUI) Enterprise. We have built strong partnerships with leading power companies like PLN, NGCP, and CGE; telecom giants such as Nokia, Ericsson, and SK Telecom; and top automotive brands including Volkswagen, Bosch, and NGK.

Certifications

Our Factory

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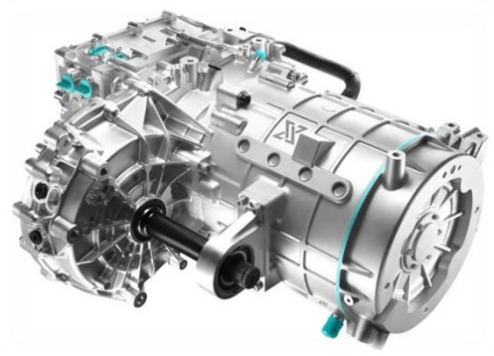
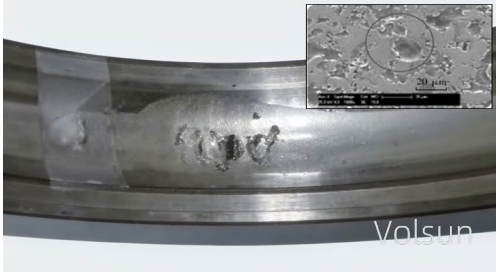
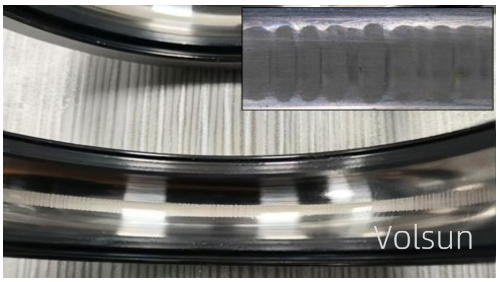
01 Bearing Electrical Erosion







Current Needs

Addressing the hazards of motor shaft electrical erosion.

Industry Status

With the rapid development of new energy vehicles, electric motors are trending toward higher voltage, higher frequency, higher speed, and higher power. The issue of bearing electrical corrosion has become increasingly severe and is an urgent challenge for all automotive companies in the industry.

Technical Status

A variety of anti-electrical corrosion technologies have matured. Based on the level of shaft electrical erosion risk, multiple measures can be reasonably combined, generally categorized into three methods: conduction, blocking, and limiting.

02 Product Range

RD Conductive Ring
ST Conductive Ring *(Segmented Ring)*

Riveted joint, suitable for new energy vehicle motors, rail transportation, industrial motors, and other applications.

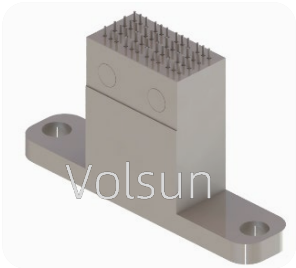
Installation methods: Press-fitting, screws, others

Granted Patents

RDW Conductive Ring
STW Conductive Ring *(Segmented Ring)*

Welded joint, designed with a modular lightweight structure, with a thickness as low as 4.5 mm. Also suitable for electrical corrosion protection in various types of motors.

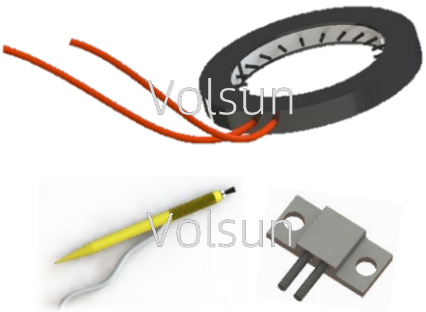
Installation methods: Press-fitting, screws, others



TDB Conductive Brush

Suitable for large-shaft-diameter motors such as wind power motors and industrial motors, often installed in the axial direction.

Installation methods: Screws, others



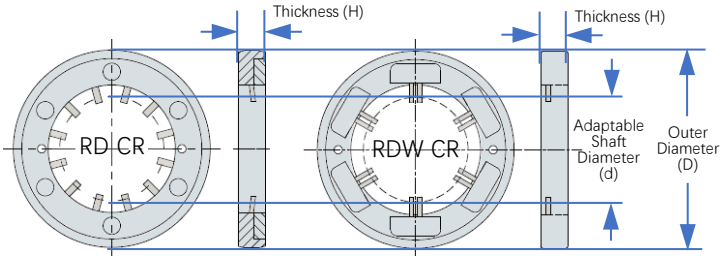
TF Testing Accessories

Testing accessories including test rings, test probes, and conductive seats, suitable for various shaft electrical testing requirements.

03 Product Selection Chart

For Water-Cooled Motors

(Dry Cavity Applications)

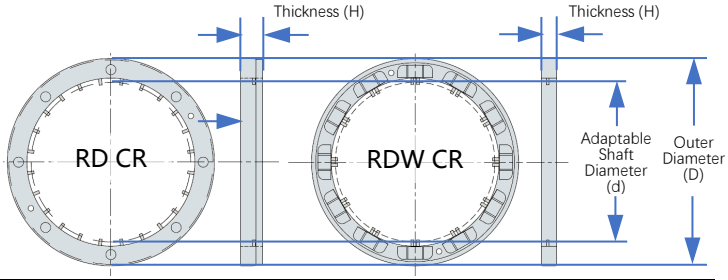


No.	Outer Diameter (D) mm	Adaptable Shaft Diameter (d) mm	Thickness (H) mm	No.	Outer Diameter (D) mm	Adaptable Shaft Diameter (d) mm	Thickness (H) mm
1	25	10	≤7.5	26	70	44.9	≤7.5
2	32	15	≤7.5	27	72	43	≤7.5
3	35	17	≤7.5	28	73	53	≤7.5
4	38	12.5	≤7.5	29	73	55	≤7.5
5	42	18	≤7.5	30	74	50	≤7.5
6	47	24.9	≤7.5	31	74	52	≤7.5
7	50	22	≤7.5	32	75	52	≤7.5
8	50.5	29.5	≤7.5	33	76	40	≤7.5
9	52	25	≤7.5	34	77	55	≤7.5
10	52	28	≤7.5	35	78	52	≤7.5
11	52.7	33	≤7.5	36	79.7	50	≤7.5
12	52.8	30	≤7.5	37	80	54	≤7.5
13	53	30	≤7.5	38	80	55	≤7.5
14	54	30	≤7.5	39	82	59	≤7.5
15	55	35	≤7.5	40	84	58	≤7.5
16	57	30	≤7.5	41	85	59	≤7.5
17	58	36	≤7.5	42	85	60	≤7.5
18	60	35	≤7.5	43	85	65	≤7.5
19	60	38	≤7.5	44	86	64	≤7.5
20	60	39	≤7.5	45	92	70	≤7.5
21	60	40	≤7.5	46	95	70	≤7.5
22	62	35	≤7.5	47	102	70	≤7.5
23	63	38	≤7.5	48	102	75	≤7.5
24	67.45	50	≤7.5	49	106	85	≤7.5
25	70	40	≤7.5	50			

Note: Dimensions in the table are for reference only. The recommended number of fiber bundles is 6–16. Custom sizing is supported. Please feel free to contact us for further inquiries.

For Oil-Cooled Motors

(Oil-Filled Cavity Applications)



No.	Outer Diameter (D) mm	Adaptable Shaft Diameter (d) mm	Thickness (H) mm	No.	Outer Diameter (D) mm	Adaptable Shaft Diameter (d) mm	Thickness (H) mm
1	38	18	≤8	31	68	35	≤7.5
2	47	24.9	≤7.5	32	69	38	≤7.5
3	47	25	≤7.5	33	70	39.7	≤7.5
4	48.5	26	≤7.5	34	70	40	≤8
5	50	25	≤7.5	35	70	46	≤7.5
6	52	26	≤8	36	70	48	≤7.5
7	52.7	29	≤7.5	37	70	50	≤7.5
8	52.7	33	≤7.5	38	70	53	≤7.5
9	52.8	30	≤7.5	39	72	50	≤7.5
10	53	25	≤7.5	40	73	55	≤7.5
11	53	30	≤7.5	41	74	49	≤7.5
12	55	29.5	≤7.5	42	74	49.8	≤7.5
13	55	35	≤7.5	43	74	50	≤7.5
14	57	35	≤7.5	44	74	53	≤7.5
15	58	39	≤7.5	45	74	57	≤8
16	58	40	≤7.5	46	75	55	≤7.5
17	60	35	≤7.5	47	78	52.7	≤7.5
18	60	37	≤7.5	48	78	54	≤7.5
19	60	39	≤7.5	49	78	55	≤7.5
20	60	40	≤7.5	50	78	60	≤7.5
21	62	35	≤7.5	51	78.6	60	≤8
22	62	38.9	≤7.5	52	78.6	60.5	≤7.5
23	62	40	≤7.5	53	80	59	≤8
24	62	42	≤7.5	54	81	58	≤7.5
25	62	42.5	≤7.5	55	82	60.5	≤7.5
26	62	43	≤7.5	56	84	61	≤7.5
27	65	33	≤7.5	57	85	65	≤7.5
28	65	40	≤7.5	58	86	58	≤7.5
29	65	45	≤7.5	59	86	65.8	≤7.5
30	66	41.5	≤7.5	60	86	66	≤7.5

Note: Dimensions in the table are for reference only. The recommended number of fiber bundles is 16–24. Custom sizing is supported. Feel free to contact us for further inquiries.

03 Product Selection Chart (Large Size and Special Types)

For Water-cooled Motors

(Dry Cavity Applications)

No.	Outer Diameter (D) mm	Adaptable Shaft Diameter (d) mm	Thickness (H) mm
1	108	75	≤7.5
2	108	79.5	≤7.5
3	108	80	≤8
4	112	61	≤8
5	115	85	≤7.5
6	120	95	≤7.5
7	121	90	≤7.5
8	122	90	≤7.5
9	128	78.5	≤7.5
10	130	95	≤7.5
11	130	100	≤7.5
12	130	105	≤7.5
13	133	95	≤7.5
14	135	110	≤7.5
15	140	85	≤10
16	142	100	≤9
17	145	95	≤7.5
18	183	130	≤7.5
19	183	145	≤15
20	200	170	≤8
21	218	180	≤15
22	345	180	≤15

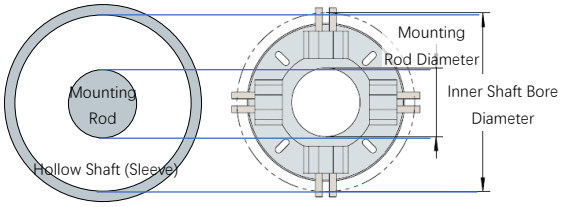
For Oil-Cooled Motors

(Oil-Filled Cavity Applications)

No.	Outer Diameter (D) mm	Adaptable Shaft Diameter (d) mm	Thickness (H) mm
1	88	55	≤7.5
2	88	63	≤7.5
3	88	67	≤7.5
4	90	55	≤7.5
5	90	65	≤7.5
6	90	66	≤7.5
7	92	72	≤7.5
8	94	75	≤7.5
9	98	55	≤7.5
10	110	55	≤10
11	114.5	51	≤11

Internally Fixed Type

(Installation Inside Hollow Shafts)



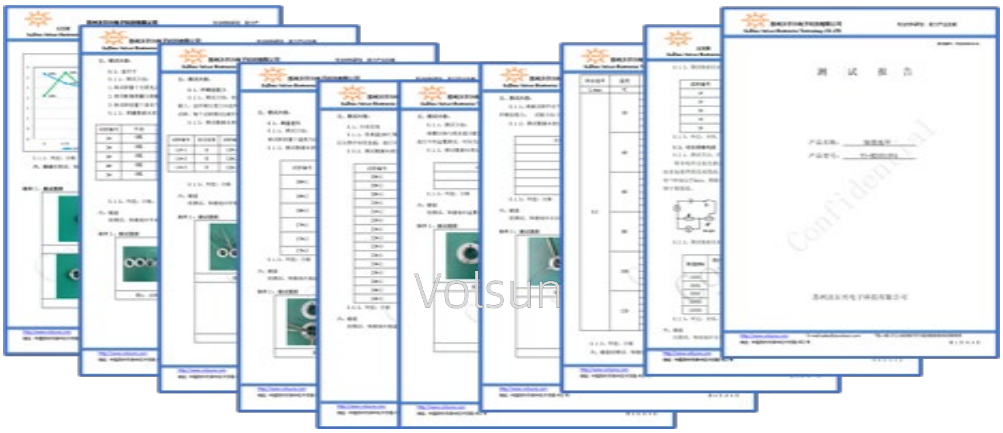
No.	Mounting Rod Diameter mm	Inner Shaft Bore Diameter mm	Thickness (H) mm
1	10	26	≤7.5

Note: Blue indicates the segmented ring design; green represents the internally fixed design with external fiber bundles.

Volsun CRs (CBs) support customized sizing. Please feel free to call us for more information.

04 Test Items

No.	Test Item	No.	Test Item
1	Prohibited Substances	9	Low Temperature Storage Test
2	Static Contact Resistance	10	High Temperature Storage Test
3	Fiber Pull-Off Force	11	Random Vibration Test
4	Dynamic Contact Resistance	12	Endurance Wear Test
5	High-Velocity Impact	13	Cleanliness Test
6	Free Fall	14	Shaft Voltage Drop Test
7	Thermal Cycling	15	Oil Resistance Test
8	Damp Heat Cycling	16	Salt Spray Test



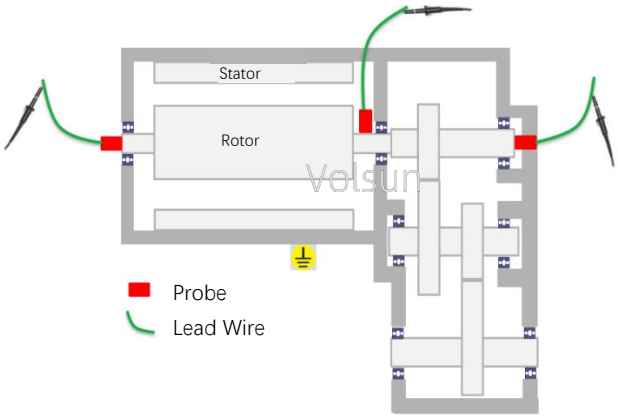
05 Shaft Voltage Test

Test Requirements:

1) Insulation resistance between the probe and housing ⩾ 1 MΩ

2) Static contact resistance between the probe and shaft < 3 Ω

3) Lead wire resistance should be < 0.2 Ω; Oscilloscope bandwidth ⩾ 200 MHz



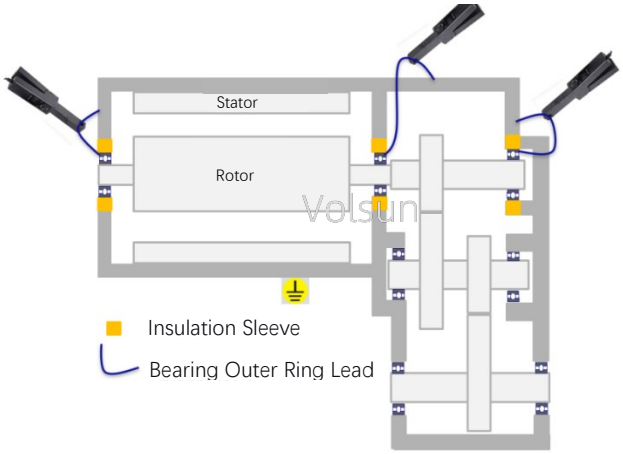
06 Shaft Current Test

Test Requirements:

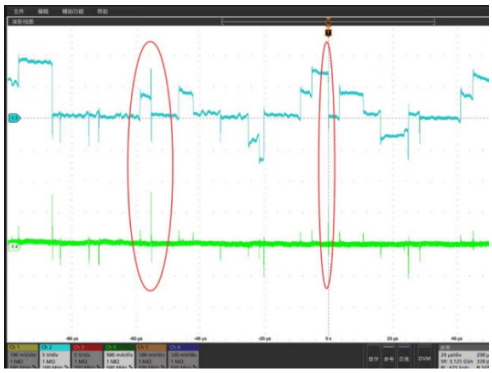
1) Insulation resistance between bearing outer ring and housing ⩾ 1 MΩ

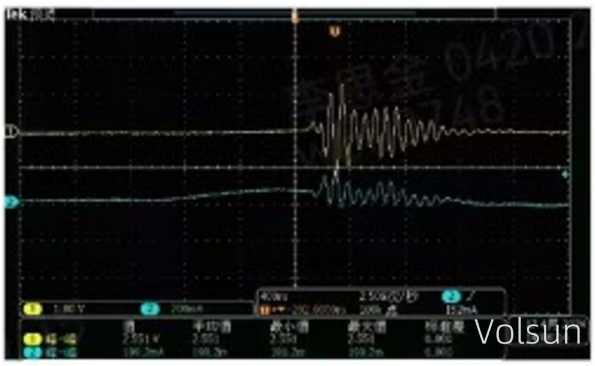
2) Lead wire resistance should be < 0.2 Ω, and lead length should follow the "minimum ground loop method" for testing

3) Current probe bandwidth ⩾ 100 MHz, accuracy 0.001 A



Typical Waveforms of Shaft Voltage/Current





Note: Testing services are available through Volsun.

WhatsApp

Address: No.402, Mudong Road, Wuzhong District,
Suzhou City, Jiangsu Province, 215101, CHINA.

Tel.: +86-18963667177 (Joe Lu)

Website: https://www.volsungroup.com/

Email: joelu@szvolsun.com